Database Management Portal for Breast Cancer Screening

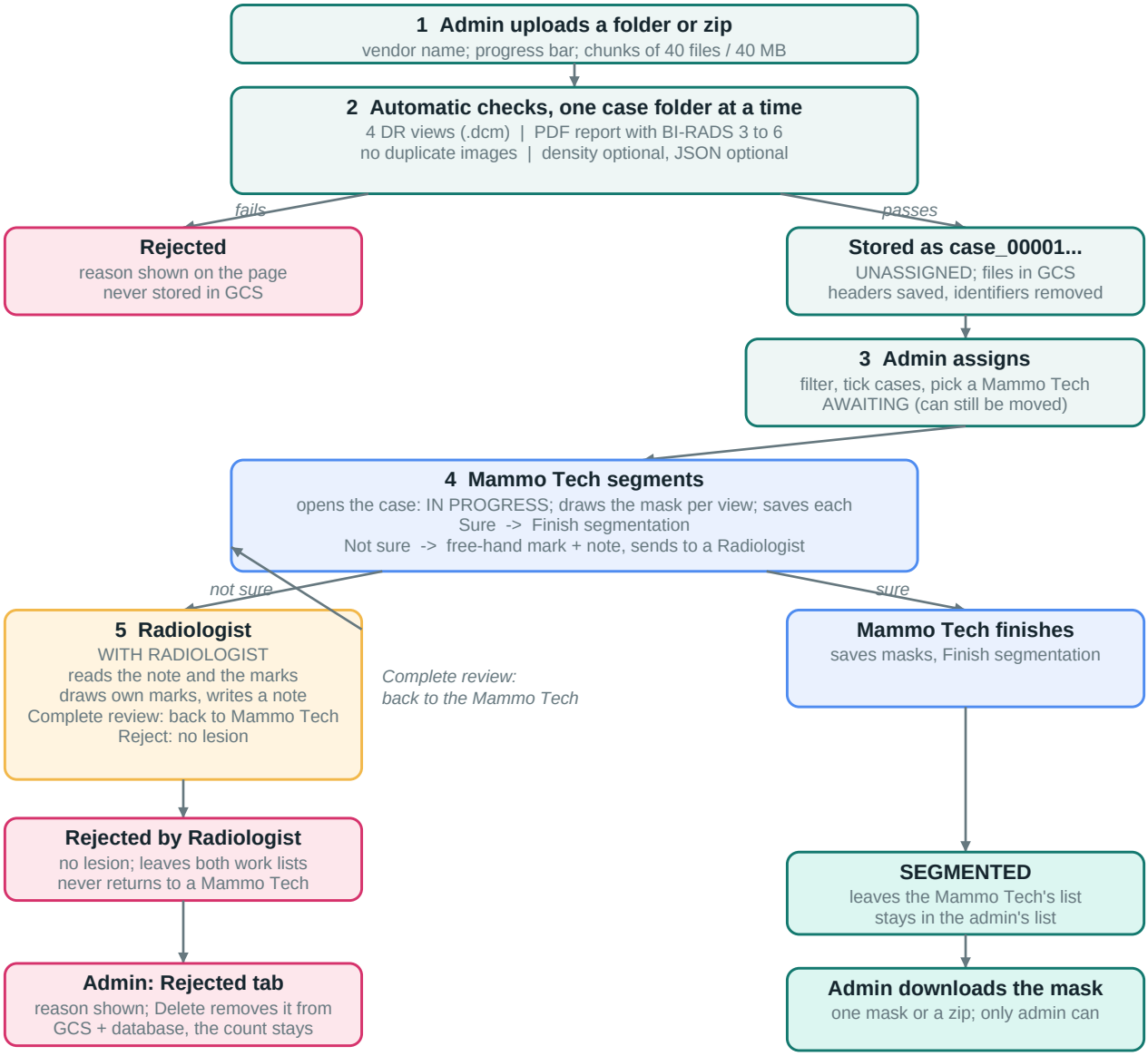
Retrospective Study

Full workflow: upload, checks, assignment, segmentation, review, hand-over and operations

Role	What they do in this module
Admin	Uploads cases, assigns them to Mammo Techs, watches progress, downloads the final masks, deletes cases a Radiologist rejected. Sees everything but never edits a mask.
Mammo Tech	Sees only the cases given to them. Segments the lesion, saves one mask per view, finishes the case. If not sure, sends it to a Radiologist.
Radiologist	Only sees cases a Mammo Tech was unsure about. Marks where the lesion may be and writes a note, or rejects the case. Does not segment.

Everything in this document describes what is built and deployed. Prospective study and the master Dashboard are separate and covered at the end only as pending work.

1. The complete flow



2. Case status: who acts next

Status	Meaning	Who acts	Next
Unassigned	Passed all checks, stored, nobody has it yet	Admin	Assign to a Mammo Tech
Awaiting	Assigned, the Mammo Tech has not opened it	Mammo Tech (admin can still move it)	Open and segment; or admin reassigns / unassigns
In progress	Mammo Tech is segmenting (also after a Radiologist hands it back)	Mammo Tech	Finish segmentation, or Not sure
With radiologist	Mammo Tech was not sure; waiting for the Radiologist	Radiologist	Open it
Radiologist in progress	Radiologist opened it	Radiologist	Complete review (back to Mammo Tech) or Reject
Segmented	Mammo Tech finished; masks saved	Admin	Download the masks
Rejected by Radiologist	No lesion; never returns to a Mammo Tech	Admin	Delete (the count stays on the dashboard)

The "not sure" loop has no limit: a case can go back and forth between the Mammo Tech and the Radiologist as many times as needed. A Mammo Tech who is sure never involves a Radiologist.

3. Upload and automatic checks

Folder layout. Any folder that holds DICOM files together with a report is one case, at any depth. Sub-folders such as DICOM, PDF, JSON or REPORT are fine, and loose files at the top (a README or a CSV) are ignored. Example:

```
omb_15_annotation_packet/
  README.txt   qc_exclusions.csv        <- ignored
  omb_15_001/
    DICOM/   RCC.dcm  LCC.dcm  RMLO.dcm  LMLO.dcm
    PDF/     report.pdf
    JSON/    report.json              <- optional
```

Check	Rule
Four views	RCC, RMLO, LCC, LMLO, each a .dcm / .DCM file. The view comes from the DICOM tags, or from the file or folder name (for example cc_left, right_mlo).
Digital Mammogram (DR)	Modality MG with a digital mammography image class. CR files, tomosynthesis and synthetic (C-view / S-view) images are rejected.
Report	A readable PDF with text (scanned pictures without text are rejected). A JSON on its own is not enough.
BI-RADS	The PDF must state BI-RADS 3, 4 (4A/B/C), 5 or 6. If left and right differ the higher one is used.
Density	A to D, taken from the PDF, otherwise from the JSON. Optional.
Duplicates	Rejected when the four images are identical to a case already stored, or to one an admin deleted earlier.
Privacy	Patient identifiers are removed from the stored header copy. A random session id is created.

A case that fails any rule is rejected with the exact reasons on the upload page and is never copied to GCS.

How the upload behaves

- The browser sends the files in chunks (up to 40 files or 40 MB each) and shows a progress bar. A large zip is sent in 40 MB parts.

- After the last chunk the server checks the cases one by one in the background; the page shows the live count of checked, passed and rejected.
- Recommended: about 25 cases per upload. The server needs free disk space about the size of the upload (a zip needs roughly double while it is unpacked).
- If the server restarts or the upload stops, the page reports the batch as interrupted. Cases already stored are kept; uploading the folder again rejects those as duplicates.
- A Google Drive download may arrive as several zips; upload each one separately with the same vendor name.

4. Screens

Admin

- **Dashboard** (admin only): Prospective numbers from the master CSV beside live Retrospective numbers.
- **Retrospective**: upload, live tally and pipeline, and the case table (case, original name, modality, BI-RADS, density, manufacturer, vendor, uploaded date and time, assigned to, status, annotation). Filters: status, BI-RADS, vendor, manufacturer, search.
- **Assigning (one way only)**: filter, tick cases, choose the Mammo Tech in the bar that appears, confirm. Select all that match a filter is available. Cases not yet opened can be reassigned or unassigned; opened cases show a lock. There is a 10-second undo.
- **View a case** (read only): the four mammograms fitted to their boxes, with wheel zoom, drag, Fit and 1:1 on each image; the report; DICOM headers; files with their folder paths; the status timeline and notes.
- **Rejected by Radiologist** (own sidebar tab): the reason, who rejected it and when. View and Delete only; nothing can be downloaded. Delete removes the case from GCS and the database; the rejected count stays.
- **QC Admin**: create Admin, Mammo Tech and Radiologist accounts. Creating a Mammo Tech no longer assigns cases; assignment is done from Retrospective.

Mammo Tech and Radiologist workspace

- **List**: only their own open cases. A finished case leaves a Mammo Tech's list; a rejected case leaves both lists. No dashboard, no history, no download of anything.
- **Layout**: choose 1, 2 or 4 boxes. Drag RCC, RMLO, LCC or LMLO from the left panel into a box. Each box has its own zoom, pan, scale bar and R / L markers.
- **Mammo Tech tools**: Pan/zoom, Paintbrush, Polygon, Magnetic lasso, Grow region, Add / Subtract, label colour and opacity, window presets, Undo / Redo, Clear mask, Save mask. Polygon and lasso have Complete, Undo last point and Clear outline; Grow region has Accept and Cancel. Click a selected tool again, or press Esc, to put it down.
- **Free hand** (both roles): Pen, Circle, Arrow with colours and thickness. Marks are only for the conversation between Mammo Tech and Radiologist, they save by themselves, carry the author's name, and are never part of the mask. The other role's marks can be removed so a new set can be drawn.
- **Notes tab**: write the note for the other role (kept on that computer until sent). The latest note from the other role is shown at the top; the history is not.
- **Side panel**: Hide it to give the images the full width; a thin strip brings it back.
- **Mask files**: one .nii.gz per view, same size as the DICOM; saved only with Save mask.

5. Emails

Sent through the same SMTP settings as every other QC email, from templates stored in the database. A mail that cannot be sent never blocks the action; the failure is only logged.

When	To	Template
Admin assigns or moves cases	the Mammo Tech	`retro_cases_assigned`
Mammo Tech sends a case as not sure	the Radiologist with the fewest open cases	`retro_case_escalated`
Radiologist completes the review	the Mammo Tech	`retro_case_returned`

Not sent yet (optional, decided for later): an email to the admin when a Radiologist rejects a case, and to the previous Mammo Tech when a case is reassigned.

6. Where things are stored

What	Where
Case files	GCS: retrospective/case_xxxxx/<VIEW>/, report/, and segmentation/ for masks. New uploads go to the bucket the server is set to (now breast-cancer-image-dataset-anonymized). Every stored file keeps its own full gs:// address, so older files in another bucket still open.
Database (qc_bcd_portal)	qc_retro_batches, qc_retro_cases, qc_retro_files, qc_retro_rejections, qc_retro_assignments, qc_retro_masks, qc_retro_notes, qc_retro_marks, qc_retro_deleted_cases; the three email templates in qc_email_templates.
Deleted cases	qc_retro_deleted_cases keeps one small record per deleted case (counts, and the image fingerprint so the same images cannot be uploaded again).
Temporary files on the server	A cache of decoded images and pictures so views open quickly; it is trimmed automatically.

7. Database changes, in the order they are run by hand

File	Purpose
`20261028_add_retro_intake.sql`	batches, cases, files, rejections
`20261029_add_retro_assignments.sql`	assignments and masks
`20261030_retro_radiologist_notes_masks.sql`	Radiologist columns, notes, per-view masks, source paths
`20261031_add_retro_email_templates.sql`	the three email templates
`20261101_retro_radiologist_reject_and_ledger.sql`	Radiologist reject columns; deleted-case record
`20261102_add_retro_marks.sql`	free-hand marks
`20261103_update_retro_email_wording.sql`	new email wording

Never edit a file that has been run; changes go in a new file. Each file starts with USE qc_bcd_portal; so it can be run from any default schema. Helper scripts (not migrations): retro_preflight.sql (read only checks), retro_test_cleanup.sql (remove test uploads) and remove_mammo_tech_users.sql (switch off or delete Mammo Tech accounts).

8. Deploying and testing

Deploy

- Push the branch, open a pull request, merge it into main. The GitHub Actions workflow builds both images and restarts the app on the VM qc-bcd-portal-instance (asia-south1-b).
- Before merging: all migrations above applied; the VM's settings contain the bucket name, SMTP details and the login link; no upload is running (the deploy restarts the app).
- The backend image installs poppler (pdftotext). Without it every report would be unreadable and every case rejected.

Test checklist on the live portal

- Upload a test folder with the vendor name TEST-DELETE-ME: cases pass, rejected ones show clear reasons.
- Open a case as admin: four images fit their boxes, the report loads, headers and files show.
- Assign to a test Mammo Tech: email 1 arrives. As that Mammo Tech: segment each view, save, Finish; the case leaves their list; as admin it shows Segmented and the mask downloads.
- Second case: Mammo Tech marks and writes a note, Not sure: email 2 arrives. Radiologist sees the marks and note, adds marks and a note, Complete review: email 3 arrives. Mammo Tech removes the Radiologist's marks, segments, finishes.
- Third case: Not sure, then Radiologist rejects with a reason. It appears under Rejected by Radiologist; delete it and check the count stays.
- Run retro_test_cleanup.sql, then the gcloud lines it prints.

When something fails

Symptom	Likely cause and what to do
Images or report say Could not load (500)	The server cannot read the file. Check the server log and the bucket setting in the VM's .env. In the VM's browser terminal: sudo docker logs qc-backend --tail 300 | grep -B3 -A15 "Render failed".
Upload shows Interrupted	The server restarted or the upload stopped. Upload the folder again; stored cases are rejected as duplicates.
Every case rejected, report unreadable	pdftotext missing in the image, or the PDF is a scan without text.
Everything slow	First open of a view downloads and decodes it once; later opens use the cache. Slowness from a laptop is mostly the distance to GCP; the VM is in the same region.
A user cannot log in	Check the account is active; Create Admin works only for an admin of the Test hospital.

9. Pending

- Import of the 1,141 cases already in the older qc_retrospective inventory (not decided).
- Optional emails: admin when a Radiologist rejects; previous Mammo Tech on reassignment.
- The old Assigned Cases tab is still on the Radiologist's sidebar.
- The reminder job ignores Retrospective cases.
- Prospective study and the master Dashboard: prototype with sample data; waiting for the real master CSVs.